Econometrics I

Problem Set 4: Birth weight case study

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1. Figure 1 shows the relationship between a baby's birth weight (in ounces) and the number of cigarettes smoked by the mother (in logarithm).

   (a) What observation would, if removed, have the largest impact on the estimated slope? Indicate in the graph and justify.

   > **Solution:** The one in the upper-left corner (around 4.5, 140). It is an'horizontal outlier' (technically, a high leverage point) that, by being largely above the regression line, must be pushing right-to-left slope up.

   (b) What observation would, if removed, have the largest impact on the estimated constant? Indicate in the graph and justify.

   > **Solution:** The one around (6.8, 50). It has no great influence on the slope because it is very close to the average $\bar{x}$. Its effect is absorbed by a lower constant.

   (c) How would the observations be distributed if the logarithm were not applied to the x-axis? Draw a graph representing the resulting relationship.

   > **Solution:** The high values of x will appear very distant from the origin and from each other in relative terms. The x values close to the origin will appear comparatively collapsed horizontally. (See figure)

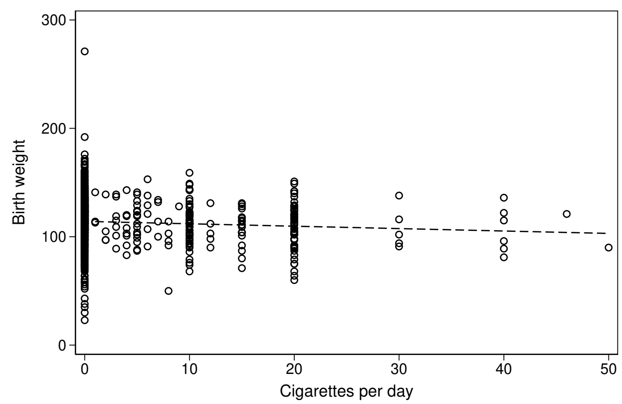

Figure 1: Birth weight vs. maternal cigarette consumption

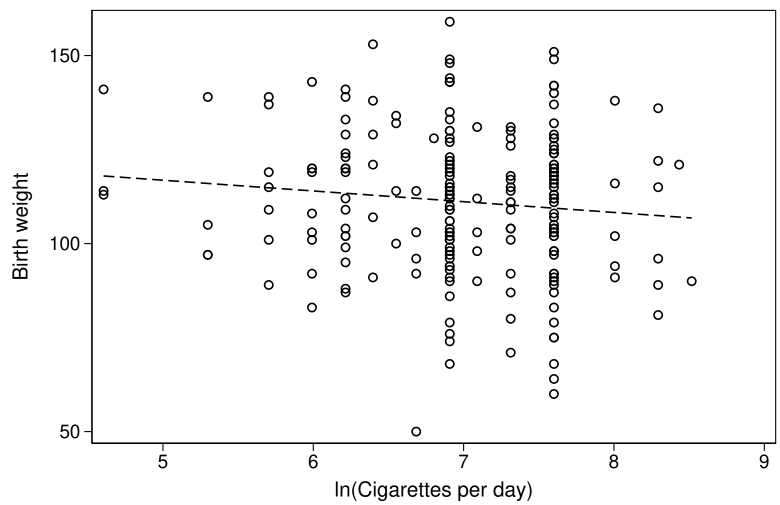

Table 1 shows a series of estimates of the effect of maternal tobacco consumption on newborn weight. Answer the following related questions.

2. How does the computer calculate the coefficients in table 1? Indicate the meaning of each symbol you use.

Solution: Each of the coefficients is obtained using $\hat{\beta}_{(k\times 1)} = (X'X)^{-1}X'y$. X is the matrix of regressors and y is the vector of the dependent variable in the database.

3. How does the computer calculate the standard errors in table 1? Indicate the meaning of each symbol you use.

Table 1: OLS Regressions

Variable dep.: Birth weight (in ounces)

	(1)	(2)	(3)	(4)	(5)	(6)
Constant	117.0***	115.4***	115.1***	118.1***	138.1***	138.5***
	(1.049)	(3.107)	(2.906)	(3.500)	(18.76)	(16.27)
Cigarettes per day	-0.463***	-0.486***	-0.582***	-0.589***		
	(0.0916)	(0.0926)	(0.109)	(0.111)		
Household income	0.0928**			0.0538	0.169?	0.202*
	(0.0292)			(0.0367)	(0.106)	(0.0922)
Mother's education years		0.331		-0.438	-1.172	-1.102
		(0.233)		(0.320)	(0.889)	(0.786)
Father's education years			0.413	0.494	1.156	
			(0.213)	(0.283)	(0.752)	
ln(Cigarettes per day)					-4.426*	-2.680
					(2.098)	(1.833)
Observations	1388	1387	1192	1191	161	212
R^2	0.030	0.024	0.030	0.033	0.070	0.036

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Solution: They correspond to the square roots of the k diagonal elements of $\frac{\hat{u}'\hat{u}}{n-k}(X'X)^{-1}$. $\hat{u}$ is the vector of residuals, k is the number of regressors contained in X, n is the number of observations.

4. What is the expected weight of a baby when the mother does not smoke and the household has an income level equal to 30?

Solution: $\hat{y} = 117 + 0.0928 \times 30$

5. What does the note with $p < 0.05$ refer to? What does p mean? Explain extensively and be very precise.

Solution: The p is the tail probability of a two-sided t-test with $n - k$ degrees of freedom under the null hypothesis that the given coefficient is zero and under all the assumption of the classical model. For example, for the constant in (1), p is the probability in repeated sampling of obtaining a value equal to or greater than 117 or equal to or less than -117, if the classical assumptions are satisfied and if the constant in the DGP is zero. The $p < 1\%$, indicates that the 99% confidence interval around 0 is contained within ± 117. Alternatively, it tells us that the 99%

confidence interval around 117 is does not contain zero whin it.

6. Any star in the household income coefficient in (5) has been replaced by a question mark. What number (if any) of stars should be on it?

Solution: The reported values tell us that the t-statistic is slightly below 1.69 ($= 0.169/0.106 \approx 1.6$). The lowest significance level with stars is 5%. With high $n-k$ (say 100), this significance level is reach with a t-statistic above 1.96. Lower degrees of freedom require an even larger t-statistic. Thus, the coefficient deserves no star.

7. Which of the estimated constants has a higher level of significance?

Solution: One must calculate $|t| = |\frac{\hat{\beta}_j}{\hat{s}_j}|$. At first glance, it is recognized that the largest t is that of column (1).

8. In table 2 (see last page) a value appears with the symbol '???'. What is the corresponding value?

Solution: It corresponds to 0.0916^2.

9. Indicate which of the following violations of assumptions would have an invalidating effect on the calculation of table 2:

   (a) Regressors correlated with the error term.

   Solution: Yes, because $E[u|X] = 0$ is no longer satisfied. With this, even unbiasedness is invalidated, which is necessary to arrive at $\sigma^2(X'X)^{-1}$.

   (b) Autocorrelation of errors.

   Solution: Yes, it invalidates because it is necessary to arrive at the term $\sigma^2(X'X)^{-1}$.

   (c) Non-normal distribution of errors.

   Solution: Does not invalidate the variance-covariance matrix.

10. Approximately, what is the 95% confidence interval of the impact of daily cigarettes according to (4)? Consider all the information given in the table.

Solution: We have a high degree of freedom $n-k(=1191-5)$. Using the asymptotic approximation of 1.96 standard deviations: CI $= -0.589 \pm 0.111$.

11. How does the impact of tobacco consumption differ between regressions (4) and (5)?

Solution: In (4), the consumption of one more daily cigarette by the mother is associated with a baby weight loss of 0.58 ounces. In (5), a 1% increase in daily cigarette consumption by the mother is associated with a baby weight loss of 0.044 ounces.

12. Why does household income appear as a significant variable in (1) but not significant in (4)?

Solution: Because income must be collinear with the education level of the parents.

13. (5) has a better R^2 than (4). Nevertheless, the coefficient associated with smoking is more significant in (4). How can this apparently contradictory difference be understood?

Solution: The line is better fitted, but regression (5) is performed with almost 10 times fewer observations, which reduces precision, reflected in larger standard errors and less significance.

14. In (4), household income and years of education of both mother and father appear non-significant. Does that mean that these variables don't matter? How could you see if they matter jointly? State the null hypothesis in matrix terms and indicate what the test statistic would be.

Solution: No, because they can be jointly significant. This can be verified using an F test based on the linear restriction $\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. The test statistic to evaluate is $\hat{F} = (R\hat{\beta} - r)'[\hat{\sigma}^2 R(X'X)^{-1}R']^{-1}(R\hat{\beta} - r) \sim F_{3,1191-5}$ with

$q = 3$ and $\hat{\sigma}^2 = \frac{\hat{u}'\hat{u}}{n-k}$.

Table 2: Variance-covariance matrix of $\hat{\beta}$ from regression 1 of table 1

Constant	Cigarettes per day	Household income	
1.1004	−0.030930	−0.025694	Constant
	???	0.00046254	Cigarettes per day
		0.00085193	Household income